

Characterization and comparative analysis of antimicrobial resistance in *Escherichia coli* from hospital and municipal wastewater treatment plants

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ABSTRACT

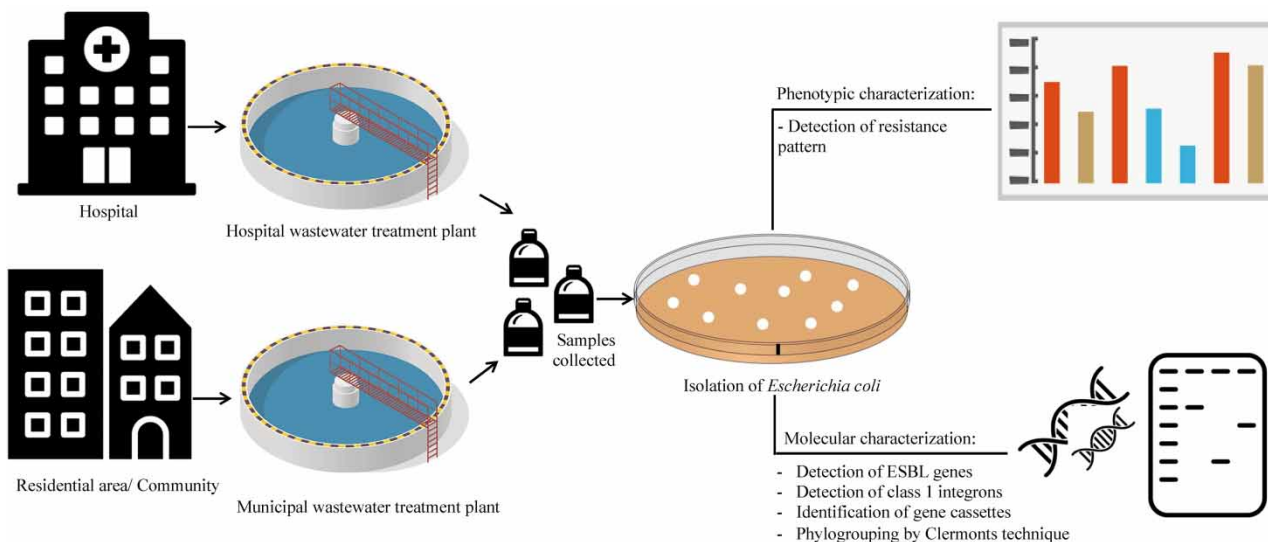
The spread of antimicrobial resistance (AMR) poses global health threats, with wastewater treatment plants (WWTPs) as hotspots for its development. Horizontal gene transfer facilitates acquisition of resistance genes, particularly through integrons in *Escherichia coli*. Our study investigates *E. coli* isolates from hospital and municipal WWTPs, focusing on integrons, their temporal correlation and phenotypic and molecular characterization of AMR. Samples from hospital and municipal WWTPs were collected over two seasons, pre-monsoon (March–May) and post-monsoon (December–February). From the hospital (hWWTP) and municipal (mWWTP) influents, 45 and 172 *E. coli* isolates were obtained, respectively. *E. coli* from hWWTP exhibited significantly higher resistance rates than mWWTP to most tested antimicrobials except tetracycline. The hWWTP isolates showed a higher prevalence (86.7%) of multidrug resistance (MDR) compared with mWWTP (48.3%). The proportion of MDR isolates from mWWTP nearly doubled in the post-monsoon season. Integron positivity was 17.7% (hWWTP) and 19.7% (mWWTP) with common gene cassettes conferring resistance to trimethoprim and aminoglycosides. Phylogroup analysis showed a predominance of group A in hWWTP and group B1 in mWWTP. The study highlights the role of hospital and municipal wastewater in disseminating AMR, with high rates of MDR *E. coli* and class 1 integrons detected.

Key words: antimicrobial resistance, *Escherichia coli*, integrons, phylogroups, wastewater treatment plant

HIGHLIGHTS

- Resistance rates of *E. coli* were higher in hWWTP compared with those from mWWTP.
- Approximately 86.7% of *E. coli* from hWWTP samples were MDR.
- MDR *E. coli* were higher in the post-monsoon sample from hWWTP and mWWTP, reflecting seasonal fluctuations.
- Approximately 19.3% of isolates had class 1 integrons.
- Spread of AMR through wastewater poses public health concerns in developing countries.

GRAPHICAL ABSTRACT



INTRODUCTION

The dynamic process of the evolution and establishment of antimicrobial resistance (AMR) is a growing concern to global public health. Research has shown that the problem of AMR requires a multipronged solution that includes understanding its evolution and spread in the environment (Hassoun-Kheir *et al.* 2020). Water systems link hospitals, communities, industries, and animal and agricultural farms from which waste generated converges in wastewater treatment plants (WWTPs). They are recognized as hotspots for the emergence of antibiotic resistant bacteria (ARB), resistance genes (ARGs), and mobile genetic elements (MGEs) as the discharge of residual antibiotics into these WWTPs can increase the selection pressure (Korzeniewska *et al.* 2013; Asfaw 2018). Hospitals are prolific users of antimicrobials and contribute significantly to the burden of AMR. While antimicrobial consumption in hospitals can be monitored, it remains unchecked in the broader community, posing challenges in tracking resistance trends (Gaşpar *et al.* 2021). Sampling wastewaters receiving sewage from specific locations offers insight into the patterns and mechanisms of resistance.

Acquisition of AMR mainly occurs through horizontal gene transfer (HGT) of ARGs, which is facilitated by discrete genetic elements such as plasmids, transposons, bacteriophages, insertion sequences (IS), and integrons. Although not mobile by themselves, integrons are commonly associated with MGEs such as transposons enabling movement of ARGs between and within species (Kaushik *et al.* 2018). The most prevalent class 1 integron is found predominantly in Gram-negative bacteria such as *Escherichia coli*, due to which their presence in the environment is a serious concern (Kaushik *et al.* 2018). *E. coli* is a well-studied commensal bacterium indicative of faecal contamination and plays a crucial role in transferring ARGs to pathogens (Shamsizadeh *et al.* 2021).

In the current study, we explored the detection and characterization of *E. coli* from hospital and municipal WWTPs with an emphasis on MGEs and their temporal correlation.

METHODOLOGY

Study location

The experiments for the study were conducted at the Department of Microbiology, Kasturba Medical College, Manipal, India, which is associated with a large 2,032 bedded tertiary care teaching hospital. As of the last census conducted, the study location of Udupi has a population of approximately 1,200,000 (2011). The atmospheric conditions of the region are of the tropical climate characterized by excessive humidity (78%) during the greater part of the year. The region also experiences significant precipitation (June–October), with average annual rainfall of 4,702 mm. The maximum temperature recorded in summer (March–May) is 38 °C, with April being the hottest month (India Meteorological Department).

The current study was focused on two key locations, as follows: (1) hWWTP, which largely receives wastewater from the main hospital (13°22′06.8″N, 74°47′16.4″E) and (2) mWWTP, which receives wastewater from various sectors of Udipi including agricultural land, non-cultivable land, built-up urban and semi-urban areas, and small-scale animal farms (13°21′32.7″N, 74°44′33.6″E) (Supplementary Figure S1). There are no significant industrial sources contributing wastewater to this plant. The capacities of these treatment plants are 300,000 and 1,200,000 litres per day (L/d), respectively, and are independent of each other. Both hWWTP and mWWTP use the aerobic activated sludge process, where wastewater passes through a bar screen chamber, followed by primary treatment (equalization tank, aeration tank, clarifier) and secondary treatment (sand filtration, activated charcoal filtration) before being chlorinated and released. Treated water from the hWWTP is repurposed for arboriculture, whereas from the mWWTP, it is released into the sea.

Sample collection

Composite samples were collected from the influent after the wastewater passes through the bar screen chamber, and from the effluent after chlorination, of hWWTP and mWWTP. Over three consecutive days, the samples were collected at three different times each day: 150 mL in the morning (8–9 am), 150 mL in the afternoon (1–2 pm), and 150 mL in the evening (5–6 pm). This amounted to 450 mL collected daily. By the end of 3 days, a total of 1.35 L were collected as samples. Additionally, to account for seasonality, the samples were collected in the pre-monsoon (March–May 2022) and post-monsoon (December 2022–February 2023) seasons. All samples were collected in sterile glass containers and transported back to the laboratory where they were maintained at between 2 and 8 °C until further processing. Approximately 1 L of the sample was used for recording basic physicochemical properties such as pH, biochemical oxygen demand (BOD), chemical oxygen demand (COD), total dissolved solids (TDS), and chloride content of all samples using [APHA \(2005\)](#) methods.

Sample processing

Out of the total 1.35 L of wastewater sample collected from each location, 150 mL was filtered through a 0.22 µm glass fibre membrane, following the [EPA \(2002\)](#) guidelines. The membrane was subsequently suspended in 10 mL of sterile physiological saline and gently vortexed to dislodge the microbial cells. The resulting suspension was then serially diluted in sterile physiological saline. From each dilution, 100 µL was spread onto Eosine Methylene Blue (EMB) (HiMedia Laboratories, India) agar plates using the spread plate technique ([Mustafa et al. 2022](#)). The plates were incubated at 37 °C for 16–18 h. After incubation, colonies displaying a characteristic shiny green metallic sheen were selected and further purified by subculturing each isolate on MacConkey agar (HiMedia Laboratories, India). The subcultures were incubated overnight at 37 °C to ensure the isolation of pure bacterial colonies. Indole testing followed by a polymerase chain reaction (PCR) assay to detect the presence of the *uidA* gene was done to confirm the *E. coli* isolates ([Martins et al. 1993](#)). Antimicrobial susceptibility testing (AST) was performed before storing the isolates at –80 °C in 15% glycerol Trypticase Soy Broth (TSB, HiMedia Laboratories, India). DNA was extracted using the heat lysis technique and stored at –20 °C for further molecular characterization.

Phenotypic characterization

AST was performed using the Kirby Bauer disk diffusion technique against amoxicillin clavulanic acid (AMC, 20/10 µg), ampicillin (AMP, 10 µg), cefepime (CEF, 30 µg), cefoperazone sulbactam (CFS, 75/30 µg), ciprofloxacin (CIP, 5 µg), ceftriaxone (CTR, 30 µg), gentamicin (GEN, 10 µg), meropenem (MEM, 10 µg), piperacillin tazobactam (PIT, 100/10 µg), trimethoprim sulphamethoxazole (SXT, 1.25/23.75 µg), and tetracycline (TET, 30 µg). All the disks were purchased from HiMedia Laboratories, India.

Methodology and interpretation were followed according to [CLSI \(2022\)](#) guidelines. Production of extended-spectrum beta-lactamase (ESBL) was tested by the double disk method ([Drieux et al. 2008](#)). Subsequently, *bla*_{TEM}/*bla*_{SHV}/*bla*_{OXA-1} multiplex PCR and *bla*_{CTX-M} multiplex PCR for groups 1, 2, 8, 9, and 25 were performed to detect the ESBL genes, as previously elucidated ([Dallenne et al. 2010](#)). The multiple antibiotic resistant index (MARI) was calculated as the ratio of the number of antimicrobials to which the bacterial isolate was resistant to the total number of antimicrobials included in susceptibility testing ([Krumperman 1983](#)).

Molecular characterization

All isolates were screened for the presence of class 1 integrons by PCR assay as described in a previous study ([Kaushik et al. 2019](#)). A subsequent PCR assay was set up for integron-positive isolates to amplify the variable region containing gene

cassettes (GC), by targeting the flanking conserved regions (Kaushik *et al.* 2019). The resulting amplicons were purified and sequenced using the Sanger's method to identify the GCs they were carrying within the variable region (Kaushik *et al.* 2019).

All isolates were categorized into four phylogroups based on triplex PCR assay targeting *chuA*, *yjaA* genes, and the DNA fragment *TSPE4.C2*, known as the Clermont technique (Clermont *et al.* 2013). The presence or absence of these marker groups isolates into A, B1, B2, or D phylogroups. The primers and PCR parameters used have been previously described (Clermont *et al.* 2013). All the primers used in this study were obtained from Eurofins, India.

Statistical analysis

AMR was calculated and represented as a rate using the following formula:

$$\text{Resistance rate} = \frac{\text{Number of resistant isolates}}{\text{Total number of isolates tested}} \times 100$$

Comparison of resistance rates between *E. coli* isolated from hWWTP and mWWTP, and between seasons, was performed using Chi-square test or Fisher's exact test, as appropriate, and was considered significant at $p < 0.05$. All analyses were done using RStudio v 4.3.0.

RESULTS

The samples were collected over two seasons, between March and May 2022 (pre-monsoon) and between December 2022 and February 2023 (post-monsoon). Detailed records of sample location, collection frequency, and basic physicochemical parameters are provided in Supplementary Table S1. In line with India's Environment Protection Rules 1986, our study assessed treated water against established limits of the Central Pollution Board (CPCB), finding that recorded values for wastewater parameters remained within acceptable ranges.

Collectively, over the two seasons, 45 and 172 isolates of *E. coli* were obtained from the influents of hWWTP and mWWTP, respectively. However, no *E. coli* was isolated from either of the effluent samples.

AST patterns of isolates obtained from hWWTP indicate higher resistance rates to ampicillin (AMP, 93.3%), cephalosporins (CTR, 82.2% and CEF, 73.3%), fluoroquinolones (CIP, 77.8%), TET (42.2%), and carbapenems (MEM, 44.4%) and lower resistance rates to aminoglycosides (GEN, 24.4%). The mWWTP isolates displayed relatively lower resistance rates compared with hWWTP, AMP (52.9%), CTR (25.6%), CIP (25%), and CEF (22.7%). The least resistance rates in mWWTP isolates were seen against GEN (3.5%) and MEM (0.6%). Hence, the resistance trends were similar in isolates from both sources, but the resistance rates were significantly higher in hWWTP isolates (p -value < 0.01), except for tetracycline where the resistance rate was higher, but not significant (p -value = 0.092). A significant difference was also seen in the number of MDR isolates (resistant to at least 1 drug from 3 classes of antibiotics), where 39/45 (86.7%) were categorized as MDR from hWWTP and 83/172 (48.3%) from mWWTP (p -value < 0.01). These findings highlight the elevated resistance levels in hospital wastewater and have been presented in Table 1. Further analysis revealed an average MARI of 0.614 and 0.212 among hWWTP and mWWTP, respectively.

For isolates obtained from the hWWTP, the prevalence of MDR isolates remained relatively stable across the two seasons. Specifically, the prevalence was 87.5% during the pre-monsoon season and slightly decreased to 85.7% in the post-monsoon season. This indicates that there was no significant change in resistance rates for these isolates across the seasons. Furthermore, we noted that for 6 out of the 11 antibiotics tested (AMP, CTR, GEN, MEM, PIT, SXT, and TET), the increase in resistance rates in the post-monsoon season was less than 10%. This pattern suggests that the environmental conditions did not significantly influence the prevalence of MDR in hWWTP isolates. By contrast, the isolates obtained from the mWWTP exhibited a marked increase in MDR prevalence, which nearly doubled from 33.3% in the pre-monsoon season to 63.5% in the post-monsoon season (p -value < 0.01). This substantial rise indicates a significant seasonal variation in resistance rates among mWWTP isolates. Specifically, we observed that the post-monsoon isolates showed increases in resistance rates of 30–80% for several antibiotics, including AMC, AMP, CEF, CFS, CTR, and TET. These findings suggest that the post-monsoon season is associated with a notable increase in the prevalence of MDR isolates from mWWTP, possibly due to an increase in the infection rates and thereby increased consumption of antibiotics in the monsoon and post-monsoon seasons.

Among the total 81 isolates resistant to ceftriaxone (37 isolates from hWWTP and 44 isolates from mWWTP), 45 (55.5%) were ESBL producers. The most prevalent ESBL genes were identified to be CTX-M-Group 1 in these isolates (37, 82.2%)

Table 1 | Comparison of resistance rates of *E. coli* isolated from hospital (hWWTP) and municipal wastewater treatment plants (mWWTP)

			ESBL producers <i>n</i> (%)	Resistance rate of <i>E. coli</i> <i>n</i> (%)											MDR <i>n</i> (%)
				AMC	AMP	CEF	CFS	CIP	CTR	GEN	MEM	PIT	SXT	TET	
hWWTP	Influent	45	15 (33.3)	30 (66.7)	42 (93.3)	33 (73.3)	27 (60)	35 (77.8)	37 (82.2)	11 (24.4)	20 (44.4)	29 (64.4)	21 (46.7)	19 (42.2)	39 (86.7)
mWWTP	Influent	172	30 (17.4)	38 (22.1)	91 (52.9)	39 (22.7)	16 (9.3)	43 (25)	44 (25.6)	6 (3.5)	1 (0.6)	23 (13.4)	31 (18)	50 (29.1)	83 (48.3)
<i>p</i> -value			0.019*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	<0.01*	0.092	<0.01*

AMC, amoxicillin clavulanic acid; AMP, ampicillin; CEF, cefepime; CFS, cefoperazone sulbactam; CIP, ciprofloxacin; CTR, ceftriaxone; GEN, gentamicin; MEM, meropenem; PIT, piperacillin tazobactam; SXT, trimethoprim sulphamethoxazole; TET, tetracycline; MDR, multidrug resistance; ESBL, extended-spectrum beta-lactamase; hWWTP, hospital wastewater treatment plant; mWWTP, municipal wastewater treatment plant.

**p*-value significant at <0.05.

followed by TEM (15, 33.3%). However, ESBL genes were also detected in isolates that were phenotypically categorized as non-ESBL producers ($n = 36$, 44.4%), with the most prevalent being TEM (16, 44.4%) and CTX-M-Group 1 (11, 30.6%). Detailed results are provided in Figure 1.

Of the isolates obtained, 17.7% (8/45) and 19.7% (34/172) of them were found to possess class 1 integrons from hWWTP and mWWTP, respectively. As can be seen in Figure 2, the integron-positive isolates obtained from hWWTP had higher resistance rates against 8/11 antibiotics tested. Whereas in integron-positive isolates obtained from mWWTP, resistance rates were

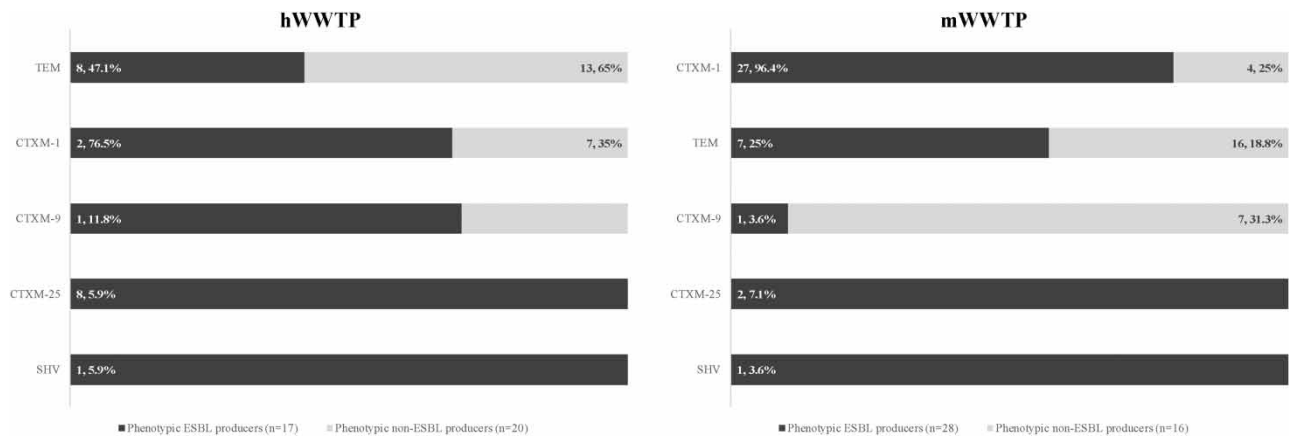


Figure 1 | Distribution of ESBL genes identified in ceftaxime resistant *E. coli* from hWWTP and mWWTP.

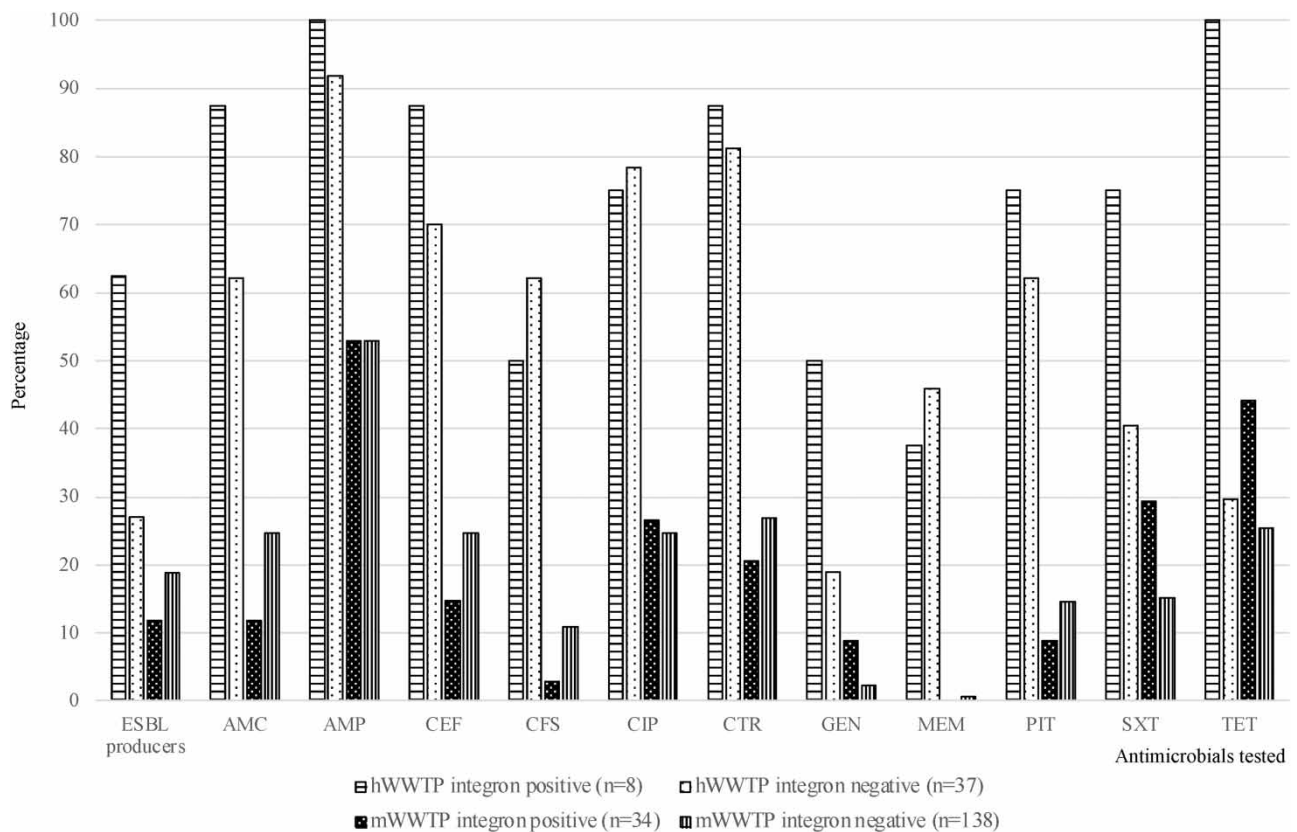


Figure 2 | Comparison of resistance rates in integron-positive and integron-negative *E. coli* isolates from hWWTP and mWWTP.

higher in only 4/11 antibiotics tested. There was no notable difference in resistance patterns between integron-positive and -negative isolates. However, it would be pertinent to note that integron-positive isolates (from both WWTPs) were consistently showing higher resistance rates against GEN and SXT. Of all the integron-positive isolates, the variable region was amplified in 26.1% (11/42). The most common GC identified was *AadA1* ($n = 3$), followed by *DfrA7* ($n = 2$). The only GC array detected was the *DfrA17-AadA5*. Single GCs such as *DfrA1*, *DfrA5*, *DfrA12*, and *AadA2* were detected only once in different isolates. GCs belonging to the *DfrA* family confer resistance to trimethoprim, and those belonging to *AadA* confer resistance to aminoglycosides such as streptomycin and spectinomycin (White *et al.* 2000; Li *et al.* 2022).

For further characterization of all isolates, Clermont multiplex PCR was performed, which revealed that more than 50% isolates were categorized into either phylogroup A or B1 from both WWTPs (Figure 3). On analysis, we observed a significant difference in the number of isolates categorized into phylogroups A and B1 ($p < 0.001$ and $p = 0.003$, respectively) between hWWTP and mWWTP, with A being more predominant in hWWTP and B1 in mWWTP. We further explored the association between resistance patterns and phylogroups (Supplementary Figure S2). Isolates that were categorized into phylogroup A showed significantly higher resistance rates to all antibiotics tested except AMP, GEN, SXT, and TET (p -value < 0.01).

DISCUSSION

Our findings underscore the critical role of hospital wastewater in driving the dissemination of AMR within the environment. A concerning 86.7% of *E. coli* isolated from these samples exhibited multidrug resistance (MDR), making wastewater a significant reservoir of resistance. Furthermore, the presence of class 1 integrons in 19.3% of these isolates highlights their contribution to resistance dissemination. While less predominant, *E. coli* isolated from municipal wastewater treatment plant also exhibited MDR, concurrently carrying integrons.

Wastewater has been recognized as a primary source of emergence of AMR in the water environment (Honda *et al.* 2023). While WWTPs act as barriers against the spread of AMR, they can paradoxically also act as reservoirs, potentially releasing elements leading to dissemination of resistance in the environment (Sambaza & Naicker 2023). Although WWTPs employ various treatment processes, such as physical, chemical, and biological techniques, to remove contaminants, inherent inefficiencies in these processes mean that ARB and ARGs are often retained in the treated water. Consequently, they may be disseminated into the environment, potentially causing untreatable or difficult-to-treat infections in humans and animals (Sambaza & Naicker 2023). WWTPs contain antibiotic residues, bacteria, genetic elements, and a rich supply of nutrients

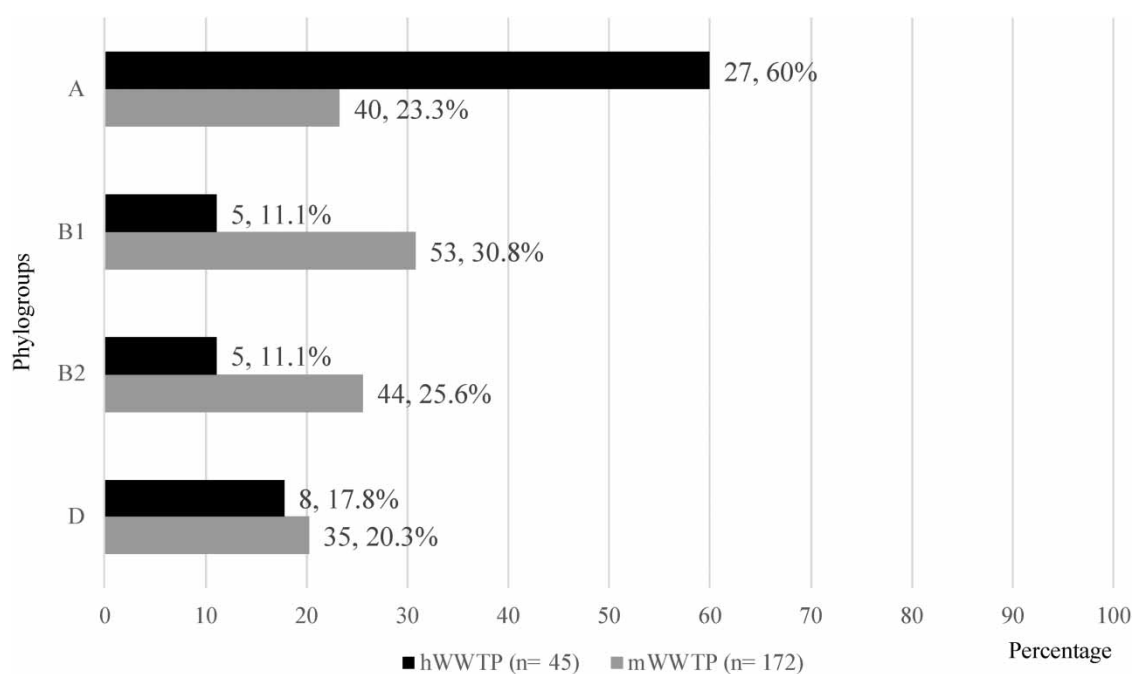


Figure 3 | Distribution of phylogroups of *E. coli* isolated from hWWTP and mWWTP.

and other factors that facilitate bacterial interactions leading to genetic transmission among bacteria (Triggiano *et al.* 2020; Sambaza & Naicker 2023). The environment within WWTPs is particularly conducive to the enrichment of the environmental resistome. Although only a portion of ARB released from WWTPs may cause disease in humans or animals, the risk of enriching the resistome through selection cannot be neglected (Manaia *et al.* 2018). A major source of AMR in wastewater is the gut microbiome, where AMR is promoted via exposure to antimicrobial doses from people in the sewershed community (Honda *et al.* 2023). While limited data are available on human microbiota owing to sampling challenges, an alternative approach is to measure antibiotic resistance in commensal indicators from untreated wastewater (Gaspar *et al.* 2021). This method allows for broader population-level sampling and serves as an early warning system for emerging resistance patterns, as initially proposed by Linton *et al.* (1974) in the 1970s and further supported by recent studies (Gaspar *et al.* 2021).

The bio-physicochemical conditions in a WWTP provide an ideal environment for bacteria. Although factors such as temperature and pH can influence bacterial communities during treatment, no specific factor reliably predicts ARB removal efficiency (Manaia *et al.* 2018). However, these factors can still affect the development of certain bacterial groups based on their characteristics (Novo *et al.* 2013; Di Cesare *et al.* 2016). Therefore, testing pH, TDS, COD, BOD, and chloride is essential for assessing wastewater quality and its impact on microbial communities and treatment processes. Monitoring these parameters helps optimize WWTP operations to control ARB proliferation and minimize environmental impact (Manaia *et al.* 2018). Chlorination as the final step in our treatment plants resulted in high chlorine content in the effluent. This could have impacted the viability of *E. coli* in effluent samples.

As mentioned previously, the samples were collected over two seasons, pre-monsoon/summer (March–May) and post-monsoon/winter (December–February). Because the monsoon season in this region witnesses very heavy rainfall, we did not collect samples as it could have resulted in dilution of our indicator organism. *E. coli* as an indicator organism is ideal as it is a common gut bacterium in humans and animals, is found in the environment, is easy to cultivate, and is well-studied for its ability to acquire resistance genes leading to spread of AMR (Zhang *et al.* 2020). For these reasons, among others, surveillance programs worldwide monitor *E. coli* to track AMR trends across human, animal, and environmental sectors, emphasizing their role in environmental research (Anjum *et al.* 2021).

While there was no difference seen in isolation frequency of *E. coli* from the influents of both sources between the seasons, there were differences observed in the MDR trends. Higher numbers of MDR *E. coli* were observed in the post-monsoon season from both hospital and municipal WWTPs (p -value < 0.01). Previous studies have illustrated seasonal fluctuations in antimicrobial consumption, indicating peak usage during winter months and reduced usage during summer (Suda *et al.* 2014; Ramsey *et al.* 2019). This pattern is consistent across various antibiotic classes, notably penicillins and cephalosporins (Ramsey *et al.* 2019). In a study conducted in India, there was a noticeable overall increase in the usage of tetracyclines, cotrimoxazole, and penicillins (typically prescribed for acute respiratory tract infections) during the winter period from December to March (Kotwani & Holloway 2011). This trend in the consumption of antibiotics was reflected in resistance rates, which was observed in our study isolates as well.

Overall, resistance rates were higher in isolates obtained from hWWTP compared with mWWTP across all antibiotics tested, with significant differences observed for 10 out of 11 antibiotics (p < 0.01). Antibiotic prescription and consumption are largely unchecked in hospitals and even in the community at large (Gaspar *et al.* 2021). However, the concentration of antibiotics and their residues released into wastewater is much higher in hospitals (Hassoun-Kheir *et al.* 2020). The exposure of microorganisms to these antibiotics may provide the necessary selection pressure for the retention of ARGs (Asfaw 2018). A systematic review by Hassoun-Kheir *et al.* (2020) referred to studies conducted in 23 different countries from 1971 to 2017 that compared resistance rates between hospital and municipal WWTPs. Out of the 37 studies included in the review, 30 studies reported results similar to ours, where resistance rates were higher in hWWTP compared with mWWTP. Gaspar *et al.* (2021) reported no significant difference in resistance rates between *E. coli* isolates from hWWTP and mWWTP, although hWWTP isolates showed slightly higher resistance rates. They concluded that community wastewater may be a substantial source of resistant bacteria, potentially comparable to the contribution from hospital wastewater sources. By contrast, our study found significantly higher resistance rates in hWWTP isolates compared with mWWTP isolates, highlighting a more pronounced role of hospital wastewater in contributing to AMR. Akiba *et al.* (2015) found that *E. coli* isolates from a WWTP handling only hospital wastewater exhibited over 80% resistance to several key antibiotics, while those from WWTPs with domestic wastewater showed significantly lower resistance of 50%, highlighting the significant impact of hospital effluent on AMR. Hospital wastewater harbours diverse ARB, ARGs, and MGEs that significantly contribute to the dissemination

of AMR, thereby emerging as environmental pollutants (Hassoun-Kheir *et al.* 2020). This poses a substantial public health concern, particularly in developing countries, where this can lead to infections in humans and animals through contaminated food, water, and direct environmental exposure. Adopting effective management practices for hospital wastewater across healthcare facilities is imperative to address this issue comprehensively (Asfaw 2018).

A rapid rise in AMR by means of production of beta-lactamase enzymes that inactivate beta-lactam antibiotics has led to the global spread of MDR strains (Islam *et al.* 2023). The CTX-M, TEM, and SHV enzymes are the primary components of ESBLs (Paterson & Bonomo 2005). TEM and SHV types likely evolved from parent enzymes such as TEM-1, TEM-2, and SHV-1 through point mutations at their active sites, expanding their activity spectrum (Islam *et al.* 2021). Additionally, there are approximately 220 variants of CTX-M-lactamases categorized into five groups, CTX-M-1, 2, 8, 9, and 25 (Naas *et al.* 2017). The most prevalent ESBLs are encoded by TEM, CTX-M, and SHV (Wang *et al.* 2022). In a study by Chandran *et al.* (2014), *E. coli* isolated from hospital wastewater was analysed for ESBL genes, revealing that 87% of the ESBL producers harboured CTX-M genes, while 63% carried TEM genes. Similarly, Siddiqui *et al.* (2018) found that *E. coli* isolated from a river exhibited 88.3 and 61.66% prevalence of CTX-M and TEM genes, respectively, among ESBL producers. They also identified the variants of CTX-M in their isolates to most commonly be *bla*_{CTX-M-15}, which belongs to CTX-M-Group 1 (Siddiqui *et al.* 2018). Our findings align with these studies, as the majority of our isolates also carry CTX-M, specifically CTX-M-Group 1 and TEM genes.

A crucial consideration is the spread of AMR, which can occur via integrons. They are capable of capturing one or more GCs that are present within the variable region and transfer them within and between species via HGT (Kaushik *et al.* 2019). Their presence in wastewaters indicates a high potential for dissemination of ARGs in that environment (Shamsizadeh *et al.* 2021). In our study, approximately 19.3% isolates were positive for class 1 integron, with higher frequency in hWWTP but not significantly different from mWWTP. When sequencing, 11 of the isolates that were carrying an integron were found to have GCs belonging to *Dfr* (conferring resistance to trimethoprim) and *Aad* families (conferring resistance to spectinomycin and streptomycin) (White *et al.* 2000; Li *et al.* 2022). Isolates that were identified to carry *Dfr* or *Aad* GCs demonstrated phenotypic resistance to trimethoprim or aminoglycosides, respectively. However, the variable region was not amplified in the majority of the integron-positive isolates. This could be due to the absence by mutation of the 3' conserved sequence, or due to the excision of GC caused by the lack of selection pressure (Oliveira-Pinto *et al.* 2017). In a study by Lamba *et al.* (2018), samples from hospital wastewater, sewer drains, and rivers in New Delhi showed that hospital effluents had the highest concentrations of total coliforms, faecal coliforms, carbapenem resistant enteric bacteria, *bla*_{NDM-1}, and class 1 integrons across summer and winter seasons. MGEs such as plasmids, integrative and conjugative elements (ICEs), transposons, IS, miniature inverted repeats, and transposable elements (MITEs) are also key players in the dissemination of AMR with known associations to a variety of ARGs (Johansson *et al.* 2023). A relatively low number of our study isolates were carrying ARGs in their class 1 integrons. A limitation was that our study was focused only on class 1 integrons; it is possible that the isolates harboured other MGEs associated with ARGs resulting in their phenotypic resistance pattern.

To further study the population genetics of our isolates, we characterized them into the four major phylogroups, namely, A, B1, B2, and D, using the Clermont triplex PCR technique (Clermont *et al.* 2013). Assigning the isolates to these phylogroups is useful for differentiating between extraintestinal pathogenic and commensal strains (Stoppe *et al.* 2017). Phylogroups A, B2, and D are most common in humans with significant geographical variations (Lagerstrom & Hadly 2023). B1 is most prevalent in animals and the environment and carries genetic factors that facilitate their adaptation to soil and water colonization (Méric *et al.* 2013). B2 and D comprise the majority of extraintestinal pathogenic strains, while A and B1 comprise intestinal pathogenic strains (Clermont *et al.* 2013; Chakraborty *et al.* 2015; Lagerstrom & Hadly 2023). While no significant differences were found, the majority of our study isolates were categorized into groups A and B1. Stoppe *et al.* (2017) combined phylogenetic data obtained from their own study and global data in a meta-analysis to identify a pattern. However, while they observed a higher frequency in groups A and D in local wastewaters, they observed that other studies in their analysis reported different findings (Stoppe *et al.* 2017). The resistance patterns observed in our study indicate that phylogroup A exhibited the highest resistance to most antibiotics, followed by phylogroup D, with phylogroups B1 and B2 showing similar resistance profiles. Our findings align with those of Chakraborty *et al.* (2015), where isolates from phylogroup B2 were the most susceptible. However, Mansour *et al.* (2024) reported the highest resistance in phylogroup B2, followed by phylogroups A and D. By contrast, our isolates demonstrated the highest resistance in phylogroups A and D, consistent with the findings of Mansour *et al.* (2024). This apparent discrepancy in the distribution of phylogroups and resistance patterns might imply that geographic location and climate could play a role in determining phylogroup distribution.

E. coli was not isolated from the effluent of either of the WWTPs in both seasons. However, the absence of their growth in laboratory conditions does not rule out the possibility of their presence. A key limitation of our study lies in its restriction to cultivable bacteria, particularly focusing on *E. coli*. While *E. coli* serves as a valuable indicator organism for AMR studies, this approach inherently excludes non-cultivable bacteria and thus may not fully represent the complexity of AMR in the sampled environments. The reliance on culture-dependent methods can overlook other important ARB and ARGs that could contribute to the broader resistome. To address this limitation, future research could incorporate advanced molecular techniques such as metagenomic sequencing. Through this technology, researchers could gain insights into the diversity, abundance, and potential transfer of resistance elements within microbial communities. This perspective could complement our current findings thus offering a more holistic understanding of AMR in environmental settings.

In wastewater treatment, addressing AMR involves destroying bacteria carrying resistance genes and degrading antibiotics within WWTPs (Sambaza & Naicker 2023). Various disinfection techniques, such as chlorination, ozonation, ultraviolet radiation, and membrane technology, have been used, although they are not specifically designed to target ARB and ARGs (Amin *et al.* 2013). Chlorination, the most common disinfection method, can produce disinfection by-products that may promote AMR through chromosome mutations (Li & Gu 2019). Membrane filtration and coagulation methods can also be employed, but they face challenges such as fouling and cost concerns (Sambaza & Naicker 2023).

Despite the growing recognition of the risks posed by ARB and ARGs, there remains a lack of decisive action to implement guidelines for monitoring wastewater. To mitigate AMR spread in wastewater treatment, policies could mandate routine monitoring of ARB and ARG levels, setting permissible limits for these elements in treated water. Regulations could also promote advanced treatment technologies, such as enhanced filtration and oxidation, specifically targeting ARB and ARGs. Financial incentives for WWTP upgrades would help address cost concerns, while stricter controls on pharmaceutical and hospital discharges into wastewater could reduce selective pressure for resistance. Encouraging interdisciplinary research would further refine these strategies, promoting sustainable AMR mitigation in wastewater systems.

CONCLUSION

In conclusion, our study underscores the critical role of hospital and municipal wastewaters in the environmental spread of AMR. The high prevalence of MDR *E. coli* and the presence of class 1 integrons in these wastewater sources highlight the significant risk posed by these environments as reservoirs of resistance. The observed seasonal fluctuation in resistance emphasizes the need for continuous surveillance of environmental isolates. The characterization of these isolates is essential to drive local policy changes alongside contributing to the global data. Continued research, including the use of advanced metagenomic sequencing, is essential to deepen our understanding of resistance dynamics and inform future interventions.

ACKNOWLEDGEMENTS

The authors thank the Manipal Academy of Higher Education, Manipal, Karnataka, for the Intramural Fund that financially supported this study. The authors also thank Ms Sukanya Shetty for all the help she has provided in the laboratory.

FUNDING

This study was supported by the Intramural Fund provided by the Manipal Academy of Higher Education, Manipal, Karnataka, India.

AUTHOR CONTRIBUTIONS

G.K. collected the samples, performed the experiments, and analysed the data. G.K. and V.K.E. drafted the paper. G.K., K.B., C.M., and V.K.E. contributed in conceptualizing and supervising the analysis of this study along with reviewing of the paper.

ETHICS APPROVAL

This study was approved by the Kasturba Medical College and Kasturba Hospital Institutional Ethics Committee with approval number 413/2021. All permissions were obtained from relevant authorities before collecting samples.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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First received 1 April 2024; accepted in revised form 11 November 2024. Available online 21 November 2024